

Fourier decay for truncated Syracuse affine laws

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7 October 2026

Abstract

Let $a_1, \dots, a_k$ be independent positive geometric random variables with $\mathbb{P}(a_i = a) = 2^{-a}$, and set $S_j = a_1 + \dots + a_j$. We study the affine law $\Phi_k(z) = \sum_{j=1}^k 3^{j-1} 2^{-S_j} + 3^k 2^{-S_k} z$ modulo 3^n , with $k \leq n$. For every fixed $A \in \mathbb{N}$, its primitive Fourier coefficients are bounded by $C_A((n-k+2)/n)^A$, uniformly in the seed and after multiplication by any unit-modulus function of S_k . Consequently, conditioning on any total-exponent slice of probability at least k^{-B} preserves decay faster than every fixed inverse power of k throughout $n-k \leq k^\beta$, for each fixed $0 < \beta < 1$. The event $S_k = 2k$ satisfies this conclusion without an additional mass hypothesis. The proof combines a fixed-dimensional Subspace Theorem obstruction to finite black-triangle covers, a qualitative renewal estimate above the entropy threshold, and a conductor recursion. The constants are ineffective. The results concern finite random affine laws and do not prove the Collatz conjecture.

1 The random affine law and the main estimates

Throughout, $\mathbb{N} = \{0, 1, 2, \dots\}$. The variables a_i are independent and take positive integer values with $\mathbb{P}(a_i = a) = 2^{-a}$. Put $S_0 = 0$ and $S_j = \sum_{i=1}^j a_i$. Since two is a unit modulo 3^n , the expression

$$\Phi_k(z) = \sum_{j=1}^k 3^{j-1} 2^{-S_j} + 3^k 2^{-S_k} z \quad \text{in } \mathbb{Z}/3^n\mathbb{Z} \quad (1)$$

is defined for every $n, k \in \mathbb{N}$ and $z \in \mathbb{Z}/3^n\mathbb{Z}$. This uses the reversed-variable convention for the Syracuse affine composition. Reversing an iid exponent list leaves both its law and its total exponent unchanged. We write

$$\chi_{n,\xi}(x) = \exp(-2\pi i \xi x / 3^n).$$

A frequency is *primitive* if $3 \nmid \xi$.

Theorem 1 (Conductor-deficit estimate). *For every $A \in \mathbb{N}$ there is a constant $C_A > 0$ such that*

$$|\mathbb{E}[\chi_{n,\xi}(\Phi_k(z))\tau(S_k)]| \leq C_A \left(\frac{n-k+2}{n} \right)^A \quad (2)$$

for all integers $n \geq 1$, $0 \leq k \leq n$, all primitive $\xi \in \mathbb{Z}/3^n\mathbb{Z}$, all $z \in \mathbb{Z}/3^n\mathbb{Z}$, and every function $\tau : \mathbb{N} \rightarrow \mathbb{C}$ with $|\tau(s)| = 1$ for all s .

The constant precedes every parameter on the right of this statement. In particular, the estimate is uniform when the deficit $d = n - k$ grows. It gives no power of n when d/n stays bounded away from zero: the exponent A is fixed, and its constant is uncontrolled as A varies.

Let $p_k(s) = \mathbb{P}(S_k = s)$. Conditioning below means the actual source law restricted to $S_k = s$ and divided by $p_k(s)$.

Theorem 2 (Conditioning on total exponent). *For each $A \in \mathbb{N}$, one constant $C_A > 0$ works simultaneously for all parameters in Theorem 1 and every s with $p_k(s) > 0$, giving*

$$|\mathbb{E}[\chi_{n,\xi}(\Phi_k(z)) \mid S_k = s]| \leq \frac{C_A}{p_k(s)} \left(\frac{n-k+2}{n} \right)^A. \quad (3)$$

Moreover, for every $q, B \in \mathbb{N}$ and $0 < \beta < 1$, there is $C_{q,B,\beta} > 0$ such that

$$|\mathbb{E}[\chi_{k+d,\xi}(\Phi_k(z)) \mid S_k = s]| \leq C_{q,B,\beta} k^{-q} \quad (4)$$

whenever $k \geq 1$, $d \in \mathbb{N}$, $d \leq k^\beta$, ξ is primitive modulo 3^{k+d} , $z \in \mathbb{Z}/3^{k+d}\mathbb{Z}$, and $p_k(s) \geq k^{-B}$.

Corollary 3 (The central slice). *For every $q \in \mathbb{N}$ and $0 < \beta < 1$, there is $C_{q,\beta} > 0$ such that*

$$|\mathbb{E}[\chi_{k+d,\xi}(\Phi_k(z)) \mid S_k = 2k]| \leq C_{q,\beta} k^{-q}$$

for every $k \geq 4$, $d \leq k^\beta$, every primitive frequency modulo 3^{k+d} , and every seed z in that group.

The proof of the corollary uses the elementary, uniform lower bound $p_k(2k) \geq 1/(4k)$. No local-limit estimate is an input.

1.1 Relation to preceding work

Tao's Proposition 1.17 gives the primitive Fourier estimate for the full-length Syracuse offset law, and his Section 7 develops the paired source and renewal geometry used here [1]. At $k = n$, the seed term vanishes modulo 3^n . We do not claim this endpoint as a new result. The distinction in (2) is its dependence on a growing conductor deficit. A bound with an exponential loss in d does not give the range $d \leq k^\beta$.

Si's Theorem H states uniform conditional decay at logarithmic oversampling [2]. Corollary 3 supplies every fixed polynomially sublinear oversampling range on the central slice; Theorem 2 covers all slices with the stated mass floor. This comparison concerns those precise parameter ranges, and does not assert a uniform theorem at a fixed positive oversampling ratio. We do not use results from Si's preprint in the proof.

2 Excluding a fixed number of black triangles

Fix $0 \leq \varepsilon < 1/4$ and a primitive frequency ξ modulo 3^n . At a lattice point $(j, l) \in \mathbb{Z}_{>0} \times \mathbb{Z}$, let

$$\Theta_{n,\xi}(j, l) = \xi 3^{2j-2} 2^{1-l} \pmod{3^n}.$$

The point is black if the absolute value of the signed representative of $\Theta_{n,\xi}(j, l)$, divided by 3^n , is at most ε . Otherwise it is white. An integer-height triangle with top-left point (j_i, T_i) and height $h_i \in \mathbb{N}$ consists of the lattice points

$$j \geq j_i, \quad l \leq T_i, \quad (j - j_i) \log 9 + (T_i - l) \log 2 \leq h_i \log 2. \quad (5)$$

Calling a triangle black means that every lattice point in it is black.

Consider $r + 1$ such triangles, indexed by $0 \leq i \leq r$, with $2(j_i - 1) < n$. Suppose there are $m_i, u_i \in \mathbb{N}$ and $e_i, g_i \in \mathbb{Z}$ satisfying

$$\begin{aligned} j_{i+1} &= j_i + m_i + u_i, & T_{i+1} - T_i &= h_{i+1} + g_i & (i < r), \\ h_i &= 4m_i + e_i, & \sum_{i=0}^r m_i &\leq n. \end{aligned} \quad (6)$$

Define the error and span by

$$E = \sum_{i=0}^r |e_i| + \sum_{i=0}^{r-1} (u_i + |g_i|), \quad H = h_0 + T_r - T_0. \quad (7)$$

The vertical gaps are signed, and no ordering of the original tops is assumed.

Proposition 4 (Finite-cover obstruction). *Fix $R \in \mathbb{N}$, $0 \leq \varepsilon < 1/4$, $\eta > 0$, and a function $w : \mathbb{N} \rightarrow \mathbb{R}$ such that, for every $\gamma > 0$, $w(n) \leq \gamma n$ eventually. There is N such that every family above with $n \geq N$, $r < R$, and $E \leq w(n)$ satisfies*

$$H \log 2 < n \log 3 + \eta n.$$

The cutoff is uniform over the primitive frequency and all triangle locations. It need not be effective.

Proof. We first treat a putative sequence of counterexamples for which every $m_i \geq \delta n$, with fixed $\delta > 0$ and fixed r . Write $q_i = 3^{n-2j_i+2}$, and let a_i be the signed representative of $\xi 2^{1-T_i}$ modulo q_i . Primitivity gives $a_i \neq 0$. Successive doublings down the left column cannot wrap while remaining black when $\varepsilon < 1/4$. Therefore

$$|a_i| \leq \varepsilon q_i 2^{-h_i}. \quad (8)$$

The height relation gives $D_i = T_{i+1} - T_i = h_{i+1} + g_i > 0$ eventually. The common frequency then gives exact integer carries z_i with

$$q_i = 9^{m_i+u_i} q_{i+1}, \quad a_i = 2^{D_i} a_{i+1} - z_i q_{i+1}. \quad (9)$$

In particular,

$$|z_i| \leq \varepsilon 2^{g_i} + \varepsilon 9^{m_i+u_i} 2^{-h_i} = \exp(o(n))$$

as an upper bound, using $e_i, g_i, u_i = o(n)$.

Put $D = \sum_{i < r} D_i$, $b = a_0$, and

$$X_0 = 2^D a_r, \quad X_{i+1} = z_i q_{i+1} 2^{D_0 + \dots + D_{i-1}} \quad (i < r).$$

Unrolling (9) gives

$$b = X_0 - \sum_{i < r} X_{i+1}, \quad 0 < \frac{|b|}{2^D} \leq \varepsilon 3^n 2^{-H}. \quad (10)$$

The single-triangle case, and the case of no nonzero carry, contradict the entropy excess directly. Otherwise pass to a subsequence with a fixed nonempty set of nonzero carry indices, and list the corresponding coordinates as $Y_1, \dots, Y_t$, in their original order.

For the integer vector $x = (X_0, Y_1, \dots, Y_t)$, use the forms $X_0 - \sum_j Y_j, Y_1, \dots, Y_t$ at infinity and the coordinate forms at both primes two and three. Each system has determinant one. Writing $|p|_p = p^{-1}$, their full local product is at most

$$\left(\prod_{z_i \neq 0} |z_i| \right) \frac{|b|}{2^D} \leq \exp(-\eta n + o(n)). \quad (11)$$

Indeed, the archimedean, two-adic and three-adic factors of a carry coordinate together contribute at most $|z_i|$; the two-adic factor of X_0 is at most 2^{-D} . All coordinates have absolute value at most $\exp(Cn)$ for a fixed $C > 0$. Taking $\kappa = \eta/(4C)$, the product in (11) is eventually smaller than $\|x\|_\infty^{-\kappa}$. The rational Subspace Theorem [3] thus places these vectors in finitely many proper rational subspaces.

None of those subspaces can contain an infinite subsequence. The exact schedules and $16 > 9$ give

$$\frac{b}{Y_j} \longrightarrow 0, \quad \frac{Y_i}{Y_j} \longrightarrow 0 \quad (i < j). \quad (12)$$

For example, the first ratio is bounded in absolute value by $(9/16)^{m_0+\dots+m_i} \exp(o(n))$ for the carry indexed by i . The ratio of successive nonzero carry coordinates has the same form with the intervening block sum. Their main exponents are linear in n , whereas all carry and scheduling losses are subexponential. If a fixed rational relation $c_0X_0 + \sum_j c_jY_j = 0$ held along an infinite subsequence, then (10) would give

$$c_0b + \sum_j (c_0 + c_j)Y_j = 0.$$

Dividing by the last coordinate with nonzero coefficient and using (12) shows that every $c_0 + c_j$ vanishes. Since $b \neq 0$, all coefficients then vanish, a contradiction.

For a general counterexample sequence, fix the number of triangles on a subsequence and make every m_i/n converge. Delete the indices with zero limit. At least one remains, since otherwise $H = o(n)$. The deleted lengths and heights total $o(n)$. Joining the schedules across each deleted segment adds these lengths to the horizontal omissions and these heights to the signed vertical gaps. The total error remains $o(n)$, and the original and retained spans differ in absolute value by $o(n)$. The retained original triangles have $m_i \geq \delta n$ for one $\delta > 0$, and retain a positive entropy excess. This reduces to the contradiction just proved. Finally, failure of a uniform cutoff would supply an unbounded counterexample sequence with the same fixed R, ε, η, w . $\square$

3 Qualitative decay above the entropy threshold

Pair the exponents and write

$$b_j = a_{2j-1} + a_{2j}, \quad L_j = b_1 + \dots + b_j.$$

The b_j are independent with $\mathbb{P}(b_j = b) = (b-1)2^{-b}$ for $b \geq 2$. For fixed pair sums, the two splits of $b_j = 3$ are equiprobable. Their contributions to the offset differ by $2 \cdot 3^{2j-2}2^{-L_j}$. Define

$$f_{n,\xi}(j, l) = \left| \frac{\chi_{n,\xi}(5 \cdot 3^{2j-2}2^{-l}) + \chi_{n,\xi}(7 \cdot 3^{2j-2}2^{-l})}{2} \right|$$

and the nonnegative envelope

$$Q_{n,\xi}(J) = \mathbb{E} \prod_{j=1}^J \begin{cases} f_{n,\xi}(j, L_j), & b_j = 3, \\ 1, & b_j \neq 3. \end{cases} \quad Q(n, J) = \sup_{3 \nmid \xi} Q_{n,\xi}(J). \quad (13)$$

Thus $0 \leq Q(n, J) \leq 1$. If $J = \lfloor k/2 \rfloor$, conditioning on the pair sums and, for odd k , the last exponent, gives

$$|\mathbb{E}[\chi_{n,\xi}(\Phi_k(z))\tau(S_k)]| \leq Q_{n,\xi}(J). \quad (14)$$

The total exponent, seed term, and terminal phase are constant during this conditional averaging. The factors from other pair sums and the unpaired exponent have absolute value at most one.

Fix a sufficiently small absolute $\varepsilon > 0$, as in the renewal estimate below. Let W_J count the indices $j \leq J$ with $b_j = 3$ for which (j, L_j) is white. The cosine formula and the definition of white give

$$Q_{n,\xi}(J) \leq \mathbb{E}e^{-\varepsilon^3 W_J} \leq \mathbb{P}(W_J \leq T) + e^{-\varepsilon^3 T} \quad (T \in \mathbb{N}). \quad (15)$$

3.1 The finite renewal estimate

The points (j, L_j) for which $b_j = 3$, in their original order, form a renewal path. Its increments are the independent blocks ending at the next occurrence of $b = 3$. Each horizontal increment is positive, each vertical increment is nonnegative, and their mean slope is four. We use the black-triangle decomposition and renewal estimate of Tao's Section 7 [1], in the following finite form.

Set $C = \lfloor n/2 \rfloor$, and use a list of $C + 1$ renewal increments. Its first increment is the starting point of a path with C subsequent steps. This list contains every raw success with horizontal coordinate at most C . The canonical black triangles cover the black points in $1 \leq j \leq C$. Starting with the first black point of this path, record its triangle; the next recorded point is the first subsequent black point above the preceding triangle's top. Continue with the same rule at path times less than C . Stops are numbered from one, so the first stop can occur at path time zero. A missing stop is distinct from a stop at zero.

Write t_R for the path time of stop R , if it exists, and let $V(t)$ count the white points at positive path times through t , restricted to $j \leq C$. The finite stopped estimate is

$$\mathbb{E} \left[\mathbf{1}_{\{t_R \text{ exists}\}} e^{-V(t_R) + \varepsilon R} \right] \leq e^\varepsilon. \quad (16)$$

It holds for every $R \geq 1$, uniformly over the initial renewal point and the primitive frequency. Averaging over the first increment therefore preserves it.

For precision, (16) uses the full cutoff C , even when the desired horizontal horizon J is smaller. It is the stopped, zero-on-absence version of the renewal argument underlying Tao's Lemma 7.9. The one-step argument uses the localized first-passage mass from Lemma 7.7 and the separation of distinct black triangles. Choose the localization radius first, then choose ε so that its separation scale contains the resulting collar. The one-step estimate applies uniformly at the restart point. We give the finite induction explicitly to specify its cutoff convention.

Let G_R denote the random variable in (16). Partition according to the first-entry event $A_{p,x}$, with first stop at path time p , point x , and canonical triangle Δ . Let $K \geq 1$ be the first additional time above the top of Δ , and U its endpoint. Write $w(U) = e^{-1}$ when U is white and lies in $j \leq C$, and $w(U) = 1$ otherwise. For $R \geq 2$, the stopping rule and white counts give

$$G_R \leq e^{-\max(p-1,0)+\varepsilon} w(U) G_{R-1}(U; \text{fresh tail}) \quad \text{on } A_{p,x}. \quad (17)$$

If $p + K \geq C$, the left side is zero. On the other branch, there are at least $\max(p-1, 0)$ positive-time white points before the first entry, and the first-exit point supplies the factor $w(U)$. The restarted variable uses a fresh list of C increments with the same absolute horizontal cutoff C . It does not replace that cutoff by $C - p - K$. Appending independent increments is legitimate because the original event and statistic depend on the original finite prefix only; the first-passage split then leaves an independent fresh tail.

Here is the geometric input for the exit weight. For an entry at depth s below its triangle's top, the relative first-passage endpoint (r, l) has probability at least $1/2$ in the window

$$|r - s/4| < L\sqrt{1+s}, \quad s < l < s + L,$$

for one absolute L , by the localized first-passage estimate. Put $a = \log 2 / \log 9 - 1/4 > 0$ and $K_L = 1 + a + L^2/(4a)$. This window lies within distance $\sqrt{K_L^2 + L^2}$ of a top-row point of the starting triangle. The canonical geometry gives separation at least $\rho = \frac{1}{10} \log(1/\varepsilon)$ between distinct triangles and a right boundary at most $n/2 - \rho$ for every triangle. Choose ε small enough that $K_L^2 + L^2 \leq \rho^2$ and $\varepsilon \leq e^{-10}$. Then the localized exit is outside its old triangle, inside the horizontal cutoff, and in no other black triangle. It is white. Consequently

$$\mathbb{E} w(U) \leq c := \frac{1 + e^{-1}}{2} \leq e^{-\varepsilon}.$$

Assuming (16) at $R - 1$, uniformly in the restart point, conditional expectation in (17) yields

$$\mathbb{E}[\mathbf{1}_{A_{p,x}} G_R] \leq \mathbb{P}(A_{p,x}) e^{-\max(p-1,0)+2\varepsilon} c \leq \mathbb{P}(A_{p,x}) e^\varepsilon.$$

Sum over the countable first-entry events. Absence of a first entry contributes zero; at $R = 1$ the pointwise bound is $G_1 \leq e^\varepsilon$. This proves (16), including an entry at time zero.

Let $E_R(J)$ be the event that stop R exists and its horizontal coordinate is at most J . On $E_R(J)$, every white point counted by $V(t_R)$ is among those counted by W_J . Markov's inequality in (16) consequently gives

$$\mathbb{P}(E_R(J) \cap \{W_J \leq T\}) \leq e^{T+\varepsilon-\varepsilon R}. \quad (18)$$

This deduction requires no uniform estimate for a process restarted with only J units of horizontal room.

3.2 Good paths cannot use only finitely many triangles

Take

$$H_n = \lceil \sqrt{n} \rceil, \quad V_n = H_n, \quad D_n = n^{3/4}.$$

Call the renewal path good if its starting horizontal coordinate is at most H_n , its horizontal and vertical steps are at most H_n, V_n , and $|l - 4j| \leq D_n$ at every path point with $j \leq J$. Exponential-moment bounds for the independent blocks and for the raw geometric sums give, for $1 \leq 2J \leq n$,

$$\mathbb{P}(\text{path is not good}) \leq B(n) := 15(n+1)e^{-\log(21/20)\sqrt{n}} + 2ne^{-\sqrt{n}/32} = o(1). \quad (19)$$

These constants follow from the increment bound

$$\mathbb{P}(\Delta j > H \text{ or } \Delta l > V) \leq 15e^{-\log(21/20)\min(H,V)}$$

and the union bound over the first $2J$ raw exponent sums, with individual tail bound

$$2 \exp \left[-\min \left(\frac{D^2}{32(2J)}, \frac{D}{8} \right) \right].$$

Fix R, T , and assume the path is good, $W_J \leq T$, and $E_R(J)$ fails. The recorded black triangles before the first crossing of J then form a family with at most R members. There is a member once n is large: before encountering the first black point, at most T white renewal points can advance the path by at most $(T+1)H_n = o(n)$.

Write (x_i, y_i) , $0 \leq i \leq r$, for the recorded entry points, and A_i for the corresponding original triangle tops. Re-anchor each triangle at its entry, using the heights

$$h_i = A_i - y_i \quad (i < r), \quad h_r = \min\{A_r - y_r, 4(J - x_r)\}, \quad T_i = y_i + h_i.$$

The new triangle with left edge x_i , top T_i , and height h_i is contained in the original black triangle: add its defining inequality to the inequality expressing membership of the entry point in the original triangle. Set

$$\begin{aligned} m_i &= x_{i+1} - x_i \quad (i < r), & m_r &= J - x_r, & u_i &= 0, \\ e_i &= h_i - 4m_i, & g_i &= y_{i+1} - T_i \quad (i < r). \end{aligned}$$

These choices satisfy (6) exactly. Only points before the crossing were assumed to lie in the tube. If (j_-, l_-) , (j_+, l_+) straddle the crossing, use the terminal height $S = \min(4J, l_+)$. Positivity of

vertical steps and $j_+ - j_- \leq H_n$ put this height within $D_n + 4H_n$ of $4J$. Put $D' = D_n + 4H_n$ to account for this endpoint.

The white gaps use at most T steps, with one exit overshoot for each intervening triangle. Thus

$$\sum_{i < r} |g_i| \leq (T + r)V_n, \quad x_0 \leq (T + 1)H_n, \quad A_r \geq S - (T + 1)V_n.$$

The tube identities and the chosen heights give $|e_i| \leq 2D' + |g_i|$ for $i < r$, and $|e_r| \leq 2D' + (T + 1)V_n$. Consequently

$$\begin{aligned} E &\leq 2(r + 1)D' + (T + 1)V_n + 2(T + r)V_n, \\ H &\geq 4J - 4(T + 1)H_n - 2D' - (T + 1)V_n. \end{aligned} \tag{20}$$

Here H is the span from (7), distinct from the step threshold H_n . The vertical gaps remain signed, and the retained horizontal lengths sum to $J - x_0 \leq J \leq n$. Since $r < R$ and R, T are fixed, the error and span loss in (20) are $o(n)$.

Suppose now that

$$\log 3 < 2\theta \log 2, \quad \theta n \leq 2J \leq n. \tag{21}$$

Then (20) implies $H \log 2 \geq 2\theta n \log 2 - o(n)$. Proposition 4, with any fixed positive η smaller than $2\theta \log 2 - \log 3$, excludes this family for all sufficiently large n . The cutoff is uniform in J and ξ . Thus a good path with $W_J \leq T$ must satisfy $E_R(J)$, and (19) and (18) yield

$$\mathbb{P}(W_J \leq T) \leq B(n) + e^{T+\varepsilon-\varepsilon R} \tag{22}$$

for all sufficiently large n , for each fixed R, T .

Proposition 5 (Uniform qualitative contraction). *If $\log 3 < 2\theta \log 2$, then, for every $\delta > 0$, there is N such that*

$$Q(n, J) \leq \delta \quad (n \geq N, \theta n \leq 2J \leq n).$$

Proof. Choose T with $e^{-\varepsilon^3 T} < \delta/3$, then R with $e^{T+\varepsilon-\varepsilon R} < \delta/3$. Finally choose n beyond the uniform cutoff in (22) and with $B(n) < \delta/3$. Apply (15). No quantitative dependence of this cutoff on R is needed. $\square$

4 The conductor recursion

Split the product in (13) after u pairs. Conditional on the first u pair sums, its suffix has conductor $n - 2u$, length $J - u$, and primitive frequency multiplied by the unit 2^{-L_u} . Independence and the supremum over primitive frequencies imply

$$Q(n, J) \leq Q(n, u)Q(n - 2u, J - u) \quad (0 \leq u \leq J, 2u < n). \tag{23}$$

Lemma 6 (Scalar bootstrap). *Suppose $0 \leq Q(n, J) \leq 1$ on $n \geq 1$, $2J \leq n$, and (23) holds. If Proposition 5 holds for some $0 < \theta < 1$, then for every $A \in \mathbb{N}$ there is $C_A > 0$ such that*

$$Q(n, J) \leq C_A \left(\frac{n - 2J + 1}{n} \right)^A. \tag{24}$$

Proof. Put $\lambda = (1 - \theta)/2$, and obtain the qualitative contraction with $\delta = \lambda^A$. For large n , set $u = \lfloor (1 - \lambda)n/2 \rfloor$ and $n' = n - 2u$. Then

$$1 \leq u, \quad \theta n \leq 2u < n, \quad \lambda n \leq n' < \lambda n + 2.$$

Let $D = n - 2J + 1$. If $D \leq \lambda n$, then $u \leq J$, and the deficit is preserved by the recursion: $n' - 2(J - u) + 1 = D$. Strong induction on n gives

$$Q(n, J) \leq \lambda^A C_A (D/n')^A \leq C_A (D/n)^A.$$

If $D > \lambda n$, the trivial bound one suffices after taking $C_A \geq \lambda^{-A}$. The finitely many smaller conductors are also absorbed in C_A , since $D \geq 1$. $\square$

Proof of Theorem 1. Choose $\theta = 5/6$. The entropy inequality follows from $3^3 = 27 < 32 = 2^5$, and $0 < \theta < 1$. Apply Proposition 5 and Lemma 6. For $J = \lfloor k/2 \rfloor$, one has $n - 2J + 1 \leq n - k + 2$. The conclusion follows from (14) and (24). $\square$

5 Conditioned affine laws

Proof of Theorem 2. For fixed s , let $\sigma_s(t) = 1$ if $t = s$ and -1 otherwise. Both 1 and σ_s are permitted terminal phases, and

$$\mathbb{E}[\chi_{n,\xi}(\Phi_k(z)) \mathbf{1}_{\{S_k=s\}}] = \frac{1}{2} \mathbb{E}[\chi_{n,\xi}(\Phi_k(z))] + \frac{1}{2} \mathbb{E}[\chi_{n,\xi}(\Phi_k(z)) \sigma_s(S_k)].$$

The triangle inequality and Theorem 1, followed by division by $p_k(s) > 0$, prove (3) with the same constant.

For (4), choose an integer M with $(1 - \beta)M \geq q + B$. When $n = k + d$, $k \geq 1$, and $d \leq k^\beta$,

$$\frac{n - k + 2}{n} = \frac{d + 2}{k + d} \leq 3k^{\beta-1}.$$

Use (3) with exponent M and $p_k(s)^{-1} \leq k^B$. The resulting upper bound is $C_M 3^M k^{(\beta-1)M+B} \leq C_M 3^M k^{-q}$. $\square$

Proof of Corollary 3. Every positive composition of s into k parts has probability 2^{-s} , and there are $\binom{s-1}{k-1}$ such compositions. Thus, for $k \geq 1$, $s \geq 1$,

$$p_k(s) = \binom{s-1}{k-1} 2^{-s}.$$

The usual convention makes this zero for $s < k$. The elementary central-binomial bound $4^k \leq 2k \binom{2k}{k}$, together with $\binom{2k}{k} = 2 \binom{2k-1}{k-1}$, gives

$$p_k(2k) = \binom{2k-1}{k-1} 2^{-2k} \geq \frac{1}{4k}.$$

For $k \geq 4$, this is at least k^{-2} . Apply Theorem 2 with $B = 2$. $\square$

6 Formal proof and scope

The accompanying Lean sources prove the actual geometric-list law, its pair-sum envelope, the canonical renewal estimate, the arithmetic obstruction, and the conductor bootstrap. The conditioned law is defined by filtering the source probability mass function on the exact exponent sum, then mapping the affine offset; its Fourier coefficient is proved equal to the normalized slice expectation. It is not an abstract distribution assumed to have that identity.

The main theorem names, in namespace `Erdos1135.Tao`, are

- `trap_unconditional_affineTwistedExpectation_deficit_all_powers`, for the joint seed and terminal-phase estimate;
- `trap_unconditional_seededAffinePMF_deficit_all_powers`, for the seeded affine probability mass function;
- `trap_unconditional_conditionedAffinePMF_polynomial_deficit_mass_floor`, for Theorem 2’s polynomial deficit range; and
- `trap_unconditional_centralConditionedAffinePMF_polynomial_deficit`, for Corollary 3.

The finite-cover argument uses the rational Subspace Theorem from Ralf Stephan’s *Subspace-Theorems* formalization [4]. The package pins its source revision and the Lean and Mathlib versions, includes the source dependency closure, and checks theorem dependencies for unproved assumptions. The Subspace Theorem is proved in that closure; it is not introduced as a new axiom.

Automated assistants were used for proof development, source review, and preparation of this manuscript. The accompanying verification records describe those checks; they do not constitute external peer review.

These estimates concern a finite random affine model. They do not bound the exceptional orbits of the deterministic Syracuse map. In particular, Fourier decay on primitive frequencies does not by itself provide the uniform integrability or integer-occupation upper bound needed to transfer these estimates to every starting integer. Neither convergence of every Collatz orbit nor absence of nontrivial cycles is asserted here.

References

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